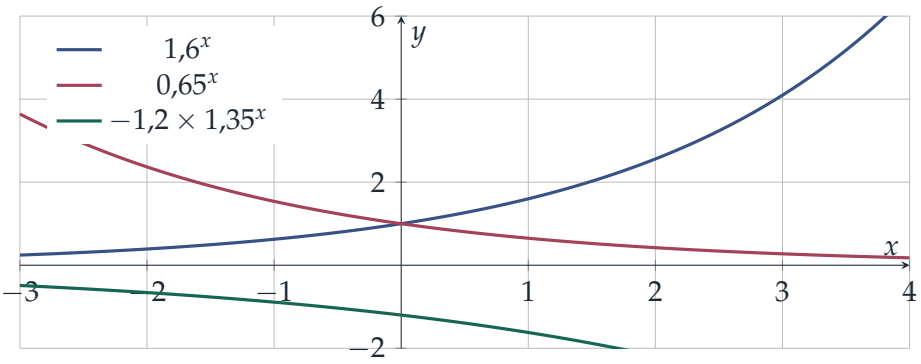
Chapter 2

Exponential functions

Skills: model a continuous change, switch representation, calculate, interpret

2.1 Reference notes

For $a > 0$, the function $x \mapsto a^x$ extends to all real numbers the integer powers of a. It satisfies $a^0 = 1$ and $a^{-x} = \dfrac{1}{a^x}$.



Key ideas

If $a > 1$, $x \mapsto a^x$ is strictly increasing. If $0 < a < 1$, it is strictly decreasing. For $f(x) = ka^x$, multiplying by $k < 0$ reverses the direction of variation. In all cases, $a^x > 0$.

For $a > 0$ and all real numbers x, y :

$$a^{x+y} = a^x a^y, \qquad a^{x-y} = \frac{a^x}{a^y}, \qquad (a^x)^y = a^{xy}.$$

Common mistake

In general, $a^x + a^y \neq a^{x+y}$. The properties concern products and quotients, not sums.

2.1.1 Changes and average rate

A quantity with initial value V_0 subject to a constant relative change with multiplier q is modelled by $V(t) = V_0 q^t$. If an overall multiplier Q results from n identical changes, the average multiplier is $q = Q^{\frac{1}{n}}$ and the average rate is $t = q - 1$.

Worked example

Turnover rises from 80,000 to 95,000 euros in four years. The overall multiplier is $Q = \dfrac{95\,000}{80\,000} = 1{,}1875$. The average annual multiplier is $q = 1{,}1875^{\frac{1}{4}} \approx 1{,}0440$: the average annual rate

is therefore approximately 4.40 %.

2.1.2 Solve and interpret

When both sides can be written with the same base, we use injectivity: for $a > 0$ and $a \neq 1$, $a^u = a^v$ is equivalent to $u = v$. Otherwise, a graphical or numerical method gives an approximate value; the common logarithm, studied later, will give a formula.

Session 7 — Definition and direction of variation of a^x

Exponential functions

06/10/2026 • B : 2 h – C : 1 h

Objectives of the session

· Give meaning to a^x.

· Link base and variation.

· Read curves.

Fluency · 6 to 8 minutes, without calculator

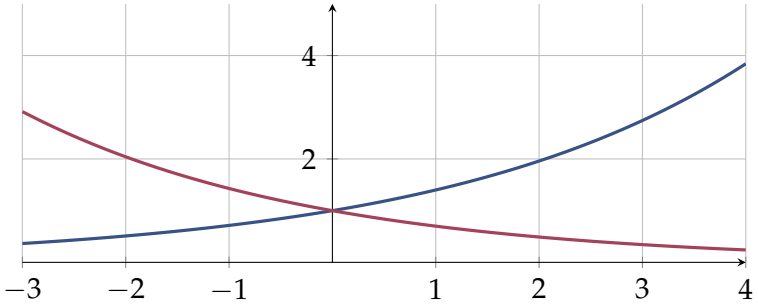
1. Write the multiplier for a 10% increase. $1 + \frac{10}{100} = 1,10$
2. Compare 1.04^3 and 1.04^7. $1,04^3 < 1,04^7$
3. Calculate $2^4 \times 2^3$. 2^7
4. Write 5^{-2} in fraction form. $\frac{1}{25}$
5. Is the function $x \mapsto 0,7^x$ increasing? **No, it is decreasing.**

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: give meaning to a^x and link the base and the variation.

Investigation

Using a calculator, complete a table of values of $1,5^x$ and $0,7^x$ for $x = -2, -1, 0, 1, 2, 3$.
Plot the points, conjecture the variations and identify the point common to both curves.



Detailed solutions · teacher edition

$a^0 = 1$, so the curves pass through $(0, 1)$. The first increases and the second decreases. Have students verbalise the piece of information that triggers the choice of method.

Written summary

For $a > 0$, a^x is defined for every real number x. The function is increasing if $a > 1$ and decreasing if $0 < a < 1$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

$a^0 = 1$, so the curves pass through $(0, 1)$. The first increases and the second decreases.

Practice · essentials

1. Simplify $3^{x+4} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 4 \times 1{,}04^x$ and $g(x) = -4 \times 0{,}8^x$.
3. An initial value of 1350 increases by 4% per year. Give the model and calculate the value after 6 years.

Detailed solutions · teacher edition

3^4 and 2^{3x}. f is increasing; $0{,}8^x$ is decreasing and the negative coefficient makes g increasing. The model is $V(t) = 1350 \times \left(1 + \frac{4}{100}\right)^t$.

References in the textbook

Extensions: Calao, lesson pp. 18-21; exercises p. 24, nos. 1-3. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 48, Exercise 2

On 1 st January 2019, France has 66,9 million inhabitants. Its annual growth rate is estimated at 0,4%. For every natural number n, let u_n be the population of France in millions on 1 st January of the year $2019 + n$ and we assume that the sequence (u_n) is a geometric sequence.

1. Give the common ratio of the sequence (u_n). Express u_n in terms of n.
2. Suggest an expression for a function f for estimating the number of inhabitants in France x years after 1 st January 2019, x being a real number. Determine, according to this model, the number of inhabitants on 1st January 2022, on 1st July 2023 and on 1 st October 2025.

Detailed solutions · teacher edition

The common ratio is 1,004 and $u_n = 66{,}9 \times 1{,}004^n$. We extend with $f(x) = 66{,}9 \times 1{,}004^x$. We calculate $f(3)$, $f(4{,}5)$ and $f(6{,}75)$, i.e. about 67.71, 68.11 and 68.72 million inhabitants.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 155 and the rate or common ratio with 3. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.

Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 8 — Powers and algebraic properties

Exponential functions

08/10/2026 • B : 1 h – C : 2 h

Objectives of the session

· Transform powers.

· Use negative exponents.

· Identify an inapplicable rule.

Fluency · 6 to 8 minutes, without a calculator

1. Simplify $x + 4 - x + 2$. $2x + 6$
2. Calculate $2^{1} \times 2^{3}$. 2^{4}
3. Write 5^{-2} in fraction form. $\frac{1}{25}$
4. Is the function $x \mapsto 0{,}7^{x}$ increasing? **No, it is decreasing.**
5. Compare 1.04^{3} and 1.04^{7}. $1{,}04^{3} < 1{,}04^{7}$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: transform powers and use negative exponents.

Investigation

Simplify $2^{3} \times 2^{-5}$, $5^{x+2} \div 5^{x}$ and $(3^{x})^{2}$. Then invent a numerical counterexample to $a^{x} + a^{y} = a^{x+y}$.

Detailed solutions · teacher edition

We get $2^{-2} = \frac{1}{4}$, $5^{2} = 25$ and 3^{2x}. With $a = 2, x = y = 1$, $2 + 2 \neq 4$? Here the equality is accidental; choose $a = 3$: $3 + 3 \neq 9$. Have students verbalise the information that triggers the choice of method.

Written summary

$a^{x+y} = a^{x}a^{y}$, $a^{x-y} = \frac{a^{x}}{a^{y}}$ and $(a^{x})^{y} = a^{xy}$. These rules require the same positive base.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

We get $2^{-2} = \frac{1}{4}$, $5^{2} = 25$ and 3^{2x}. With $a = 2, x = y = 1$, $2 + 2 \neq 4$? Here the equality is accidental; choose $a = 3$: $3 + 3 \neq 9$.

Practice · essentials

1. Simplify $3^{x+5} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 5 \times 1{,}05^x$ and $g(x) = -5 \times 0{,}8^x$.
3. An initial value of 1400 increases by 5% per year. Give the model and calculate the value after 6 years.

Detailed solutions · teacher edition

3^5 and 2^{3x}. f is increasing; $0{,}8^x$ is decreasing and the negative coefficient makes g increasing. The model is $V(t) = 1400 \times \left(1 + \frac{5}{100}\right)^t$.

References in the textbook

Extensions: Calao, lesson pp. 18-21; exercises p. 24, nos. 1-3. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 49, Exercise 10

Write as a single power of 2:

$$a = 2^5 \times 2^3; \qquad b = \left(2^{4,5}\right)^3; \qquad c = 2^3 \times 4^2; \qquad d = \frac{2^5}{4}.$$

Detailed solutions · teacher edition

$a = 2^8$, $b = 2^{13,5}$, $c = 2^3 \times (2^2)^2 = 2^7$ and $d = 2^{5-2} = 2^3$.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 160 and the rate or common ratio with 4. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.
Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.
Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 9 — Functions $x \mapsto ka^x$: variation and graph

Exponential functions

13/10/2026 • B : 2 h – C : 1 h

Objectives of the session

· Study ka^x.

· Combine sign and variation.

· Link formula and graph.

Fluency · 6 to 8 minutes, without a calculator

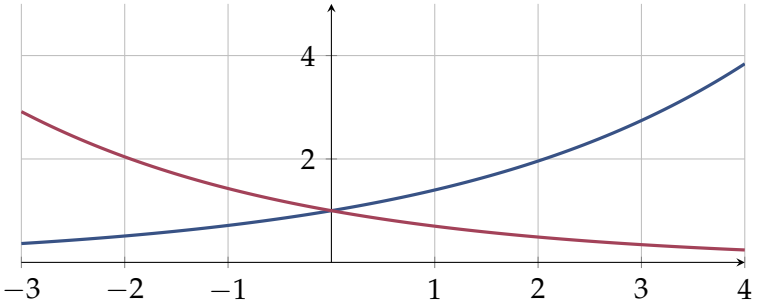
1. Calculate 20 per cent of 100. 20
2. Write 5^{-2} in fraction form. $\frac{1}{25}$
3. Is the function $x \mapsto 0{,}7^x$ increasing? **No, it is decreasing.**
4. Compare 1.04^3 and 1.04^7. $1{,}04^3 < 1{,}04^7$
5. Calculate $2^2 \times 2^3$. 2^5

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: study ka^x and combine sign and variation.

Investigation

Match the four functions 2^x, $0{,}6^x$, -2^x and $-0{,}6^x$ to four curves. Justify each choice by the sign, the y-intercept and the direction of variation.



Detailed solutions · teacher edition

All have absolute value equal to 1 at 0. The sign distinguishes the two pairs; the base and the sign of the coefficient determine the variation. Have students put into words the information that triggers the choice of method.

Written summary

For $f(x) = ka^x$, the sign is that of k. If $k < 0$, multiplication by k reverses the direction of variation of a^x.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

All have absolute value equal to 1 at 0. The sign distinguishes the two pairs; the base and the sign of the coefficient determine the direction of variation.

Practice · essentials

1. Simplify $3^{x+6} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 6 \times 1{,}06^x$ and $g(x) = -6 \times 0{,}8^x$.
3. An initial value of 1450 increases by 6% per year. Give the model and calculate the value after 6 years.

Detailed solutions · teacher edition

3^6 and 2^{3x}. f is increasing; 0 $,8^x$ is decreasing and the negative coefficient makes g increasing. The model is $V(t) = 1450 \times \left(1 + \dfrac{6}{100}\right)^t$.

References in the textbook

Extensions: Calao, lesson pp. 20-21; exercises p. 24, nos. 11-13, then p. 28, nos. 43-44. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 48, Exercise 6

Determine the direction of variation of the functions defined on $\mathbb{R}$ by:

$$a : t \mapsto -2 \times 1{,}4^t; \qquad b : t \mapsto 9{,}85 \times 0{,}95^t; \qquad c : t \mapsto 0{,}8 \times 2{,}25^t.$$

Detailed solutions · teacher edition

$1{,}4^t$ is increasing and the coefficient -2 reverses the direction of variation: a is decreasing. 0 $,95^t$ is decreasing and the coefficient is positive: b is decreasing. 2 $,25^t$ is increasing and the coefficient is positive: c is increasing.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 165 and the rate or common ratio with 5. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.

Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 10 — Average rate of change

Exponential functions

15/10/2026 • B : 1 h – C : 2 h

Objectives of the session

· Calculate an average rate.

· Distinguish overall and average rate.

· Interpret an exponent $\frac{1}{n}$.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier for a 25% increase. $1 + \frac{25}{100} = 1,25$
2. Is the function $x \mapsto 0,7^x$ increasing? **No, it is decreasing.**
3. Compare 1.04^3 and 1.04^7. $1,04^3 < 1,04^7$
4. Calculate $2^3 \times 2^3$. 2^6
5. Write 5^{-2} in fraction form. $\frac{1}{25}$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: calculate an average rate and distinguish overall and average rates.

Investigation

The price of a service increases from 18 to 23,50 euros in five years. Calculate the overall multiplier, then the average annual multiplier and the average annual rate. Check by compounding the change five times.

Detailed solutions · teacher edition

$Q = \frac{23,5}{\quad} \approx 1,3056$, $q = Q^{\frac{1}{5}} \approx 1,0548$, i.e. about 5 ,48% per year. Have students verbalise the information that triggers the choice of method.

Written summary

If Q is the overall multiplier of n identical changes, the average multiplier is $q = Q^{\frac{1}{n}}$ and the average rate is $q - 1$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

$Q = \frac{23,5}{18} \approx 1,3056$, $q = Q^{\frac{1}{5}} \approx 1.0548$, i.e. about 5.48% per year.

Practice · essentials

1. Simplify $3^{x+2} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 2 \times 1{,}02^x$ and $g(x) = -2 \times 0{,}8^x$.
3. An initial value of 1500 increases by 2% per year. Give the model and calculate the value after 6 years.

Detailed solutions · teacher edition

3^2 and 2^{3x}. f is increasing; 0 ,8^x is decreasing and the negative coefficient makes g increasing. The model is $V(t) = 1500 \times \left(1 + \frac{2}{100}\right)^t$.

References in the textbook

Extensions: Calao, lesson pp. 20-21; exercises p. 24, nos. 11-13, then p. 28, nos. 43-44. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 49, Exercise 18

A company's turnover increased by 38% over three years. Calculate the average annual increase to the nearest 0.1%.

Detailed solutions · teacher edition

The overall multiplier is 1 ,38. If t is the average annual rate, $(1+t)^3 = 1{,}38$, therefore $t = 1{,}38^{1/3} - 1 \approx 0.113$. The average annual increase is about 11.3%.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 170 and the rate or common ratio with 6. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.
Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.
Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 11 — Reading graphical representations

Exponential functions

03/11/2026 • B : 2 h – C : 1 h

Objectives of the session

· Read images and preimages.

· Compare two models.

· Estimate graphically.

Fluency · 6 to 8 minutes, without a calculator

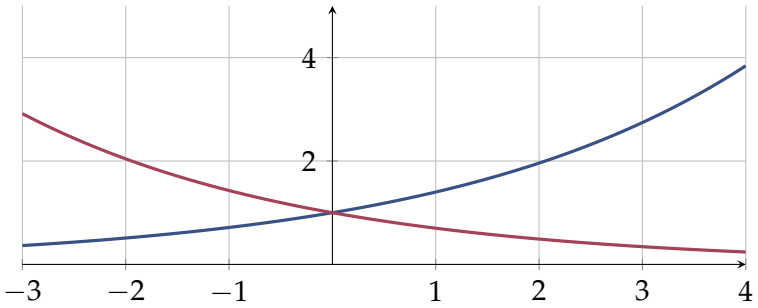
1. Simplify 3 $x + 7 - x + 2$. $2x + 9$
2. Compare 1.04 ^{3} and 1.04^{7}. $1{,}04^3 < 1{,}04^7$
3. Calculate $2^{4} \times 2^{3}$. 2^7
4. Write 5 $^{-2}$ in fraction form. $\dfrac{1}{25}$
5. Is the function $x \mapsto 0{,}7^x$ increasing? **No, it is decreasing.**

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: read images and preimages and compare two models.

Investigation

Two commercial offers are modelled by $f(x) = 120 \times 1{,}08^x$ and $g(x) = 180 \times 0{,}96^x$. Draw the curves on $[0, 12]$, read their intersection point and interpret the regions where each value is the larger.



Detailed solutions · teacher edition

Reading the graph places the intersection near $x \approx 3{,}5$. Before this value, g dominates; after, f dominates.
An approximate value is expected. Have students say aloud the piece of information that triggers the choice of method.

Written summary

Reading $f(x)$ means starting from x to the curve then to the y-axis. Finding a preimage of y means starting from y to the curve then to the x-axis.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

Reading the graph places the intersection near $x \approx 3{,}5$. Before this value, g dominates; after, f dominates. An approximate value is expected.

Practice · essentials

1. Simplify $3^{x+3} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 3 \times 1{,}03^x$ and $g(x) = -3 \times 0{,}8^x$.
3. An initial value of 1550 increases by 3% per year. Give the model and calculate the value after 6 years.

Detailed solutions · teacher edition

3^3 and 2^{3x}. f is increasing; $0{,}8^x$ is decreasing and the negative coefficient makes g increasing. The model is $V(t) = 1550 \times \left(1 + \dfrac{3}{100}\right)^t$.

References in the textbook

Extensions: Calao, p. 52, nos. 31 to 36. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 52, Exercise 38

Below are the graphs of the functions f and g defined by: $f(x) = 2{,}9^x$ and $g(x) = 0{,}35^x$.

1. What is the graph of f? Justify this choice.
2. Solve the equation graphically $f(x) = 4$. A graphing calculator can give an approximate solution to within 0.01.
3. Determine graphically the set of real numbers x such that: $f(x) \leqslant 4$ and $g(x) \leqslant 4$.

Detailed solutions · teacher edition

The curve of f is increasing because $2{,}9 > 1$; that of g is decreasing because $0 < 0{,}35 < 1$. We obtain $f(x) = 4$ for $x \approx 1{,}30$. Thus $f(x) \leqslant 4$ for $x \leqslant 1{,}30$. For g, the threshold is $x \approx -1{,}32$ and $g(x) \leqslant 4$ for $x \geqslant -1{,}32$.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 175 and the rate or common ratio with 7. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.

Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 12 — Simple exponential equations

Exponential functions

05/11/2026 • B : 1 h – C : 2 h

Objectives of the session

· Solve with the same base.

· Approximate a solution.

· Check plausibility.

Fluency · 6 to 8 minutes, without a calculator

1. Calculate 5 per cent of 160. 8
2. Calculate $2^{1} \times 2^{3}$. 2^{4}
3. Write 5^{-2} in fraction form. $\frac{1}{25}$
4. Is the function $x \mapsto 0{,}7^{x}$ increasing? **No, it is decreasing.**
5. Compare 1.04^{3} and 1.04^{7}. $1{,}04^{3} < 1{,}04^{7}$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: solve with the same base and approximate a solution.

Investigation

Solve $4^{x} = 64$, $9^{x} = -\frac{1}{}$ and $1{,}12^{x} = 2$: first by using the same base where possible, then numerically for the last one. Bound the last solution between two integers.

Detailed solutions · teacher edition

$x = 3$; $9^{x} = 3^{-1}$ so 2 $x = -1$ and $x = -\frac{1}{2}$; finally $x \approx 6{,}12$, between 6 and 7. Have students verbalise the information which triggers the choice of method.

Written summary

To solve $a^{x} = b$: use the same base when possible; otherwise bound or use a digital tool. Check that $b > 0$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

$x = 3$; $9^{x} = 3^{-1}$ so 2 $x = -1$ and $x = -\frac{1}{2}$; finally $x \approx 6.12$, between 6 and 7.

Practice · essentials

1. Simplify $3^{x+4} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 4 \times 1{,}04^x$ and $g(x) = -4 \times 0{,}8^x$.
3. An initial value of 1600 increases by 4% per year. Give the model and calculate the value after 6 years.

Detailed solutions · teacher edition

3^4 and 2^{3x}. f is increasing; $0 < 0{,}8 < 1$, $0{,}8^x$ is decreasing and the negative coefficient makes g increasing. The model is $V(t) = 1600 \times \left(1 + \frac{4}{100}\right)^t$.

References in the textbook

Extensions: Calao, p. 52, nos. 37-38, then p. 54, nos. 45-46. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 52, Exercise 33

Solve on $\mathbb{R}$ the inequality: $0.8^x \geqslant 0{,}8^3$.

Detailed solutions · teacher edition

The function $x \mapsto 0{,}8^x$ is strictly decreasing; the order of the exponents is therefore reversed. The solution is $x \leqslant 3$, i.e. $]-\infty; 3]$.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 180 and the rate or common ratio with 2. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.
Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.
Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 13 — Modelling with an exponential function

Exponential functions

10/11/2026 • B : 2 h – C : 1 h

Objectives of the session

· Build an exponential model.

· Identify the parameters.

· Predict and interpret.

Fluency · 6 to 8 minutes, without calculator

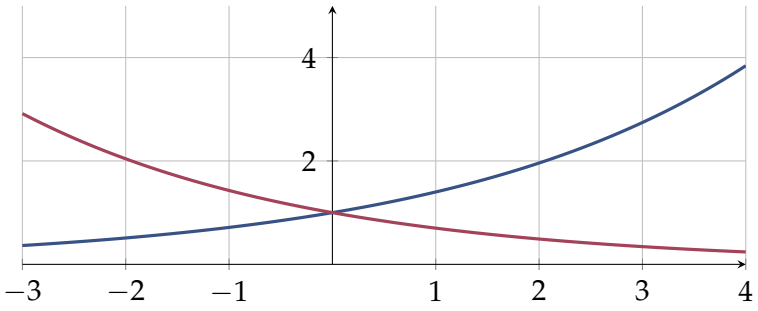
1. Write the multiplier for a 10% increase. $1 + \frac{10}{100} = 1,10$
2. Write 5^{-2} in fraction form. $\frac{1}{25}$
3. Is the function $x \mapsto 0,7^x$ increasing? **No, it is decreasing.**
4. Compare 1.04^3 and 1.04^7. $1,04^3 < 1,04^7$
5. Calculate $2^2 \times 2^3$. 2^5

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: build an exponential model and identify the parameters.

Investigation

An app has 12,000 users at launch and its usage increases by 9% per month. Construct $f(t)$, calculate $f(6)$, then determine numerically when it will exceed 25,000 users.



Detailed solutions · teacher edition

$f(t) = 12\,000 \times 1,09^t$, $f(6) \approx 20,126$. The threshold is exceeded from the ninth month on. Have students put into words the information that triggers the choice of method.

Written summary

A constant relative change with multiplier q is modelled by $f(t) = V_0 q^t$. V_0 is the initial value and q the multiplier per unit of time.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

$f(t) = 12\,000 \times 1{,}09^t$, $f(6) \approx 20{,}126$. The threshold is exceeded from the ninth month on.

Practice · essentials

1. Simplify $3^{x+5} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 5 \times 1{,}05^x$ and $g(x) = -5 \times 0{,}8^x$.
3. An initial value of 1650 increases by 5% per year. Give the model and calculate the value after 6 years.

Detailed solutions · teacher edition

3^5 and 2^{3x}. f is increasing; $0{,}8^x$ is decreasing and the negative coefficient makes g increasing. The model is $V(t) = 1650 \times \left(1 + \dfrac{5}{100}\right)^t$.

References in the textbook

Extensions: Calao, p. 50, nos. 20 to 25, then p. 53, no. 41. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the textbook's digital resources.

Calao textbook · p. 50, Exercise 25

Atmospheric pressure is 1,013 hPa (hectopascal) at sea level and decreases by 12% each time one climbs 1,000 m. This is exponential decay.
It can be modelled by a function P of altitude h in thousands of metres satisfying: $P(h) = ka^h$.

1. Determine the constants k and a.
2. Calculate the pressure at an altitude of 5,500 m, to the nearest hPa.

Detailed solutions · teacher edition

$P(0) = k = 1{,}013$ and a decrease of 12% gives $a = 0{,}88$. Thus $P(5{,}5) = 1\,013 \times 0{,}88^{5{,}5} \approx 502$ hPa.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 185 and the rate or common ratio with 3. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.

Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 14 — Chapter review and assessment

Exponential functions

12/11/2026 • B : 1 h – C : 2 h

Objectives of the session

· Use what was learnt in the chapter.

· Write up a model.

· Correct an error.

Fluency · 6 to 8 minutes, without a calculator

1. Simplify $3\quad x+3-x+2$. $2x+5$

2. Is the function $x \mapsto 0{,}7^x$ increasing? **No, it is decreasing.**

3. Compare $1.04^{\ 3}$ and 1.04^7. $1{,}04^3 < 1{,}04^7$

4. Calculate $2^{\ 4} \times 2^3$. 2^7

5. Write $5^{\ -2}$ in fraction form. $\dfrac{1}{25}$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: use what the chapter has taught and write up a model.

Investigation

From four pieces of student work, identify and correct: rate/multiplier confusion, incorrect property of powers, error in the direction of variation and conclusion without a unit.

Detailed solutions · teacher edition

The correction must name the error, write the correct rule, redo the calculation and conclude in the context. Have students verbalise the information that triggers the choice of method.

Written summary

To validate an exponential model: identify V_0 and q, calculate with units, check the order of magnitude and interpret the result.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

The correction must name the error, write the correct rule, redo the calculation and conclude in the context.

Practice · essentials

1. Simplify $3^{x+6} \div 3^{x}$ and $(2^{x})^{3}$.
2. Study the direction of variation of $f(x) = 6 \times 1{,}06^{x}$ and $g(x) = -6 \times 0{,}8^{x}$.
3. An initial value of 1700 increases by 6% per year. Give the model and calculate the value after 6 years.

Detailed solutions · teacher edition

3^{6} and 2^{3x}. f is increasing; 0 ,8^{x} is decreasing and the negative coefficient makes g increasing. The model is $V(t) = 1700 \times \left(1 + \frac{6}{100}\right)^{t}$.

References in the textbook

Extensions: Calao, p. 54, nos. 47 to 58. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 55, Exercise 59

1. A share falls by 20% one year, then by 80% the following year. What is the average annual percentage decrease?
2. What is the average rate of change corresponding to a 60% increase followed by a decrease of 60%?

Detailed solutions · teacher edition

1. The overall multiplier is 0 ,8 $\times$ 0,2 = 0,16. The average annual multiplier is $\sqrt{0{,}16} = 0{,}4$, i.e. a average decrease of 60%.
2. The overall multiplier is 1 ,6 $\times$ 0,4 = 0,64. The average multiplier is $\sqrt{0{,}64} = 0{,}8$, i.e. an average decrease of 20%.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 190 and the rate or common ratio with 4. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.
Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.